

AIS Glass — What Changed and What Every Number Means

Oct 5, 2026 · @vijji babu

A plain-English guide to the data change we just made, and to every figure on the screens — what it is, where it comes from, and what it does not mean.

What we changed

Overall Analysis now shows everything the client gave us — all 53 branches and all 2,063 products. Before, it only showed the 2 branches and 136 products we built the forecasting on, which meant the page was describing about a sixth of their business and calling it the picture.

Nothing else moved. Forecasting, Training and Per Branch & SKU still work on the 2 branches and 136 products, exactly as before. And no chart, filter or tile changed shape — it is the same page, reading a bigger pile of data.

The short version: **one page answers "what does the client's business look like", the rest answer "what did we model".** Those were always two different questions; now they get two different answers instead of one wrong one.

Why two sets of numbers, and why they don't match

If you open Overall Analysis and then Forecasting, the numbers look wildly different. That is correct, not a bug. They are counting different things.

	Overall Analysis	Forecasting, Training, Per Branch & SKU
Branches	All 53	2 (Bengaluru, Delhi-1)
Products	All 2,063	136
Orders received	₹860.04Cr	₹60.24Cr
Units ordered	2,146.8K	146.8K
What it answers	What is the client's business?	What did we actually forecast?

Each page says which it is, right at the top. Overall Analysis carries a grey **FULL NETWORK** band; the others carry a blue **WORKSPACE** band. If you ever see a number and aren't sure which world it is from, look at that band first.

Why we didn't just forecast everything. Forecasting 53 branches × 2,063 products means roughly 70,000 separate product-location combinations, each needing 13 models fitted and compared. That is days of compute, and most of those combinations sell a handful of units a year. The pilot deliberately picked 2 branches and the 136 products that carry about 70% of their value, so the modelling work lands where the money is.

Overall Analysis — every number on the page

These are the live full-network figures, covering **2025-W14 to 2026-W31 (70 weeks)**.

The three tiles at the top

Orders received — ₹860.04Cr, 2,146.8K units ordered. What customers asked for. Every order line in the client's order file, multiplied by that product's average MRP. This is demand, not sales — it counts what was wanted whether or not it was ever delivered.

Sales despatched — ₹716.94Cr, 2,134.0K units. What actually went out of the door against those orders. Same products, same prices, just counting what shipped instead of what was asked for.

Orders not despatched — ₹143.10Cr, 468.4K units, 83.4% of orders covered. The gap between the two above. ₹860.04Cr asked for minus ₹716.94Cr shipped. The 83.4% is the value version of the fill rate: for every ₹100 of orders, about ₹83 was delivered.

The numbers behind those tiles

Figure	Value	What it means
Fill rate	82.0%	Of the units ordered, 82% were despatched. Counted in units.
Fill rate by value	83.4%	The same thing in rupees. Slightly higher, so the lines that got short-shipped were cheaper than average.
Short units	504,298	Units on lines where <i>less</i> was sent than ordered. Only the under-deliveries.
Net gap	468,431	Short units <i>minus</i> over-deliveries. Smaller than the figure above, because on some lines more was sent than ordered.
Censored rows	142,526	Order lines where despatch fell short. On these, the order is a <i>floor</i> on real demand — see the glossary.

Figure	Value	What it means
Ordered-demand share	55.35%	How much of the history is real orders rather than the sales stand-in. The rest is older data from before the order book starts.
Product-location lines	68,597	Distinct branch × product combinations in the data.
Rows of history	1,052,033	Total weekly records behind the page.

Why "short units" and "net gap" are different, and why we show both. 504,298 units were under-delivered. But on other lines the branch sent *more* than was ordered, and those over-deliveries cancel some of it out, leaving a net 468,431. Reporting only the net would hide a real service problem; reporting only the gross would overstate it. Both are on the page on purpose.

The charts, and what each one is for

The page has three tabs above the charts. They are three ways of asking about the same demand.

Where demand comes from — who and what is driving orders.

- *Demand by branch × value class* — which locations order most, split by how expensive the products are. A tall bar that is mostly top-band products is a different business from a tall bar of cheap volume.
- *By glass type, by vehicle category, by vehicle age* — the same total sliced three other ways. Useful for "are we a laminated business or a toughened one", "passenger or commercial", "new vehicles or replacement".
- *By product group and by SKU* — the same again, down to the individual product.

Over time — the trend. Orders and despatches week by week (or month by month, using the grain dropdown). This is where you see seasonality and whether the gap between ordered and despatched is widening.

How well we deliver — the service view. Fill rate, shortfalls and which branches or products are worst served.

Two controls worth knowing

FROM / TO sets the window. It defaults to April 2025 because that is when the real order book starts — before that the data is the sales stand-in only, and mixing the two made every total disagree with the tiles.

By month / By week / By quarter changes the grain. These are real aggregations of the same underlying weekly rows, so switching does not change any total — only how finely it is cut. Daily is not offered, because the source data has no daily detail and splitting a week across its days would be inventing numbers.

Forecasting and Training — the modelling numbers

These pages are still on the 2 branches and 136 products. Currently **272 product-location lines have a chosen model**, forecasting **26 weeks ahead** from 2026-W31.

The four tiles on a line

Chosen model — which of the 13 models won for *this* product at *this* branch. It differs line to line, and that is the point: we fit all 13 to every line and keep whichever was most accurate there. Across the current run the winners are spread — Exponential Smoothing on 55 lines, LSTM on 40, XGBoost on 33, and so on. No single model is best everywhere.

Accuracy — how close the forecasts were on data the model had never seen. 90% means the typical miss was about 10%.

MAPE — the same thing stated as error instead of accuracy. 10% MAPE means the average forecast was off by 10%.

Horizons returned — how many weeks ahead we produced a number for, out of how many were asked for. "26 / 26" means every week came back.

Two things about accuracy worth knowing

Accuracy and MAPE are not simply 100 minus each other. They usually agree, but accuracy is floored at zero. On a line where the forecast was off by more than 100%, accuracy shows 0% while MAPE might show 357%. Both are printed side by side so the real size of the miss is visible rather than hidden behind a floor.

There is no single headline accuracy, on purpose. Every line is forecast separately, so a network-wide "accuracy" would be an average over lines that have nothing to do with each other. The pages show accuracy per line, and the median across lines — not one number pretending to describe everything.

Why the pages open on one branch and one product

There used to be "All locations" / "All SKUs" options. They quietly meant something different — a model fitted to *summed* demand across everything, which scores very differently (169% error against about 10% for a typical single line). Summing demand

smooths out exactly the lumpiness the forecast exists to handle, so that number flattered nothing and confused everyone. Both pages now open on one real line.

Page: Per Branch & SKU

Overall Analysis is the whole picture. This page is the opposite — you pick one location and one product and look at just that line. It is scoped to the 2 branches and 136 products.

The controls: Location, SKU, Value class, Grain. Pick a branch and a product and everything below is about that combination alone.

The four tiles

Tile	What it is
Ordered demand	Units customers asked for on this line, over the window
Unfilled	How much of that never went out
Fill rate	Despatched ÷ ordered, counting only lines where a despatch was recorded
Series covered	How many branch × product lines your current filter includes

The four charts

- **Month-of-year pattern** — all the Januaries together, all the Februaries together, and so on. This is where seasonality shows up. If this product always spikes in March, you see it here rather than guessing from a wiggly line.
- **Month-on-month change** — how much it moved versus the month before, as bars above and below zero. Good for spotting a step change: a product that quietly doubled, or fell off.
- **Price per unit** — average MRP over time for this line. Worth a glance because a jump in demand *value* with flat units is a price move, not a demand move.
- **Monthly detail** — the underlying table. Every period with its ordered, despatched and shortfall figures. This is the one to open when a chart looks odd and you want to see the actual rows.

"Reading this combination" is a short written note the page generates about whatever line you picked — how lumpy its demand is, whether it has enough history to model well, that sort of thing.

Page: Forecasting

Pick a location and a SKU; the page forecasts that one line 26 weeks ahead from 2026-W31.

The main chart shows real history as a solid line, then the forecast continuing from where history stops. Around the forecast are shaded bands — those are the service levels (q80, q90, q95). The band is not "how confident we are"; it is "stock to this level and you cover demand this often". The q95 edge is the one the order quantities use.

A horizon with no band is drawn as a bare point. That means we had a forecast but not enough past errors on that line to put an honest range around it — so we draw none rather than invent one.

The four tiles are the ones described earlier: chosen model, accuracy, MAPE, horizons returned.

The supporting panels

- **Why this model, for this series** — the reason this model won here. Not a generic blurb: it names what the selector measured.
- **What the model was given** — the inputs. How much history, what seasonality was detected, which extra variables were used. Worth opening when a forecast looks wrong, because the answer is usually here.
- **The six months, added up from the weeks** — a weekly forecast summed to a six-month total, so the number is comparable with how planners usually think.
- **Demand Drift** — whether recent demand has moved away from the older pattern, with a projected crossing point if it continues. The page labels this a *stated line*, not a *statistical result* — it is a straight-line projection to make the trend visible, not a second forecast.

History range lets you zoom the chart. It changes what you see, not what was forecast.

If a line has no trained model, the page says "*No trained line to forecast*" rather than drawing zeros. A flat line at zero and "we never modelled this" look identical on a chart, and they are completely different situations.

Page: Training

This is where you see the models being judged. Pick a line, press **Train all models**, and 13 models race on that line's history.

The race bars show each model's accuracy on the chosen line, best first. It is a real race — each model is fitted, then scored on data it never saw.

The leaderboard columns

Column	Meaning
#	Position
Model	Which of the 13
Accuracy	How close it got, as a percentage
MAPE	The same miss stated as error
WAPE	Total error ÷ total demand. Another angle — a model can lead on one and trail on the other
Bias	Direction of the miss. Negative means it forecasts too low , which is the direction that causes stockouts
Ran on	How many lines it completed, plus how many it was ineligible for (+) or failed on (!)
Wins	How many lines it was picked champion for

Every model stays on the board, even the ones that didn't run. A model that couldn't fit shows *Ineligible* with the exact reason — usually not enough history. Hiding it would make the field look stronger than it is.

The four baselines. Alongside the 13 we run four deliberately dumb rules: repeat last value, repeat last year's same period, 3-period average, 6-period average. They can never be champion. They exist as a sanity check: if a real model can't beat "just repeat last month", that is worth knowing, and the page says so plainly rather than quietly dropping the comparison.

The configuration panel shows the rules the run used — which metric picks the champion (MAPE), the tie-breaks, how many validation points a model needs before it is allowed to win, the per-model timeout. These are visible so a result can be argued with rather than taken on trust.

Page: Ordered vs Dispatched Time

This page asks one question: **the branch master says a lead time — is that what actually happens?**

It takes two things the client already gave us and puts them side by side. Avg Lead Time from Location Master, and Despatch Date minus Order Date from the order file. Subtracting one existing date column from another. Nothing is rewritten in either file.

Coverage tiles — how many order lines went into it, how many branches, how many need a look, and how many lines were thrown out. A line is used when the duration is between 0 and 90 days. The excluded ones are date errors, not slow deliveries: the worst has a despatch date in **2002** against a 2025 order. They are counted on screen rather than quietly dropped.

The trend chart — observed lead time by order month, a line per branch, with the master's stated figure as a dashed reference. Monthly points, not smoothed, so you can see the month something changed.

The distribution chart — how long orders actually took, as a share of lines by whole days. The mean hides the tail; this shows it. The last bar is a catch-all so the slow tail is visible rather than running off the edge.

Stated against observed, by branch — grouped bars: the master's figure, the measured mean, the measured p95. Grouped and never stacked, because stacking would imply they add up to something, and a planning assumption plus a measured duration do not.

By SKU — observed duration per product. No stated figure to compare against, because Location Master is per branch, not per product. We don't draw a comparison that doesn't exist.

The comparison table lists every branch with stated, observed, median, p95 and the gap, plus a plain-English note on anything worth reviewing.

Two findings this page produced

Four branches state a zero-day lead time and actually take 2.8–4.8 days. Rudrapur says 0, takes 4.75.

Invoice Date is not a third checkpoint. About 40% of lines are invoiced *before* despatch, clustered at exactly one or two days early. That is a billing practice, not bad data — but it means Order → Despatch → Invoice is not a sequence, so the page reports the shape and refuses to average it into a duration.

This page compares; it does not correct. Nothing here changes a forecast, a recommended order or a safety stock figure. A gap is not automatically an error — the master's number may be a deliberate planning allowance, and only your team can say.

Page: AI Recommendations — what it actually rests on

This is the one people assume is "the AI deciding". It is not. **No number on this page comes from the AI.** The figures and the ranking are computed; the AI only writes the English sentence around them.

Right now it ranks **261 product-location lines: 85 critical, 155 high, 11 medium, 10 with no recommendation.**

Where the order quantity comes from

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protection period = review period + lead time
order-up-to       = q95 forecast scaled to that period
recommended order = order-up-to - usable stock - already on order +
backorders
  
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A real line from the current run — **Bengaluru × FG.BA5.LFH.GCG2120000:**

Input	Value	Where it comes from
q95 demand	2,644 units/month	The stored forecast, champion model <code>sarimax</code>
Review period	30 days	A configured setting
Lead time	4 days	Location Master, branch average
Protection period	34 days	30 + 4
Usable stock	0	Stock snapshot, 1 Aug 2026
On order / backorders	0	Absence, not measurement — the client files contain no open-order snapshot
Recommended order	2,954 units	2,644 scaled to 34 days, minus nothing on hand

How urgency is decided

Four rules, checked in this order. No scoring, no model:

1. Couldn't produce a figure → **no recommendation** (and it says why)
2. No usable stock while there is real demand → **critical**
3. Less cover than the protection period → **high** — it will run out before a replenishment can arrive
4. Otherwise → **medium**

The order matters. Rule 3 is the real inventory logic: having less cover than the window in which no stock can arrive means the stockout is already booked.

What the AI is allowed to do

It gets the already-ranked lines and writes the explanation. It is instructed it may not do arithmetic, may not reorder or merge lines, must name the branch and full product code, and must drop anything it cannot tie to a figure it was given.

Two things enforce that, because instructions alone are not enough:

- Its text is matched back to its line **by key, never by position** — so it cannot end up describing the wrong branch.
- The computed reason is printed **above** its prose, always. Remove the AI entirely and the page is still correct.

If the model is unreachable, the figures and ranking still render with templated wording, and the card is marked *template*.

Words that keep coming up

Censored demand. An order line where we shipped less than was asked for. It matters because the recorded order is then only a *floor* on what the customer really wanted — if they'd wanted 200 and we could only offer 150, the file says 150. So on those lines, real demand is at least that, probably more. 142,526 lines across the network are like this. Any forecast built on them is built on a slight understatement, and the pages flag it rather than quietly living with it.

Sales proxy. The order book only starts in April 2025. For history before that we use invoiced sales as a stand-in for demand. It is labelled differently everywhere because it is not the same measurement — an invoice tells you what was sold, not what was asked for. Currently 55% of the history is real orders and the rest is this stand-in.

Fill rate. Of what was ordered, how much went out. 82% in units, 83.4% in rupees.

q95. When we recommend an order quantity we don't use the average forecast, we use a level that covers demand 95 times out of 100. Averages leave you short half the time. It is a *service level*, not a confidence interval — a common mix-up.

Protection period. The stretch of time an order has to cover: how long until you next place an order (30 days), plus how long delivery takes. If delivery takes 4 days, you are covering 34 days, so the order is sized for 34 days of demand.

Champion model. The one of the 13 models that scored best for a given product-location line. Chosen by measured accuracy, per line, not picked by hand.

Ordered vs Dispatched Time. The page comparing what the branch master says a lead time is against what the order dates actually show. It compares; it does not correct anything, and it does not feed the order quantities.

What these numbers don't tell you

Worth having straight before anyone asks in a meeting.

Overall Analysis counts only weeks where something happened. It is built from the actual order and sales records, not from a grid with a zero in every empty slot. Totals, percentages and trends are exact. But a count of *rows* means something different here than on the forecasting pages, so don't compare row counts across the two.

The ₹860.04Cr is demand, not revenue. It values orders at average MRP. It is what customers asked for at list price, not what was invoiced or collected.

Forecast accuracy comes from the pilot slice only. 2 branches, 136 products. It says nothing about how the models would do on the other 51 branches, and especially not on slow-moving products — those are harder to forecast, and the 136 were picked for being the opposite.

Stock figures are a single snapshot dated 1 August 2026, while demand history runs to July 2026. So anything stock-related is a current estimate, never a trend.

There is no proof that acting on the recommendations would have helped. Testing that needs stock history, and the client's files don't contain any. The recommendation pages are framed as things to look at, not instructions — deliberately.

Lead time is measured only as far as despatch. Order date to despatch date. There is no goods-receipt date in the file, so the leg from despatch to arrival is invisible to us.